1 Generalized Lagrange Equation

Author: Xianchao Date: June 30, 2026

Starting from the principle of the least action, which has the definition:

$$S[t] = \int_{t_1}^{t_2} L(\boldsymbol{x}, \dot{\boldsymbol{x}}, t)\ \mathrm{d}t \tag{1.1}$$

Here the t is the parameter, usually we use it as time. $\boldsymbol{x}$ is the generalized coordination, and L is the Lagrange. $\boldsymbol{x}$ is t dependent which means $\boldsymbol{x}(t)$. According to the principle of the least action, we have:

$$\delta S = 0 \tag{1.2}$$

We only assume the variation of x, which gives:

$$\begin{aligned}\delta S &= \int_{t_1}^{t_2} \delta L(\boldsymbol{x}, \dot{\boldsymbol{x}}, t)\ \mathrm{d}t \\ &= \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i}\delta x^i + \frac{\partial L}{\partial \dot{x}^i}\delta \dot{x}^i + \frac{\partial L}{\partial t}\delta t\ \mathrm{d}t \\ &= \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i}\delta x^i + \frac{\partial L}{\partial \dot{x}^i}\delta \dot{x}^i\ \mathrm{d}t \end{aligned} \tag{1.3}$$

And we have:

$$\begin{aligned}\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}^i}\delta \dot{x}^i\ \mathrm{d}t &= \left[\frac{\partial L}{\partial \dot{x}^i}\delta x^i\right]\Bigg|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}^i}\right)\delta x^i\ \mathrm{d}t \\ &= -\int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}^i}\right)\delta x^i\ \mathrm{d}t \end{aligned} \tag{1.4}$$

And now we have:

$$\delta S = \int_{t_1}^{t_2} \frac{\partial L}{\partial x^i}\delta x^i - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}^i}\right)\delta x^i\ \mathrm{d}t = 0 \tag{1.5}$$

Thus we have the Euler-Lagrange equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}^i}\right) - \frac{\partial L}{\partial x^i} = 0 \tag{1.6}$$

We can define $\frac{\partial L}{\partial \dot{x}^i} = p^i$, applying Legendre transformation, we have Hamiltonia:

$$H(\boldsymbol{x}, \boldsymbol{p}) = p_i\dot{x}^i - L(\boldsymbol{x}, \boldsymbol{p}) \tag{1.7}$$

Actually the term $p_i\dot{x}^i$ is the so called symplectic potential, and we can replace the $\boldsymbol{x}$ and $\boldsymbol{p}$ in a very general form $\boldsymbol{\xi}$, so the Lagrange can be written as:

$$L = -a_i(\boldsymbol{\xi})\dot{\xi}^i - H(\boldsymbol{\xi}) \tag{1.8}$$

Using this form, we can substitute it into the Euler-Lagrange equation, which has:

$$\frac{\partial L}{\partial \xi^i} = -\frac{\partial a_j}{\partial \xi^i}\dot{\xi}^j - \frac{\partial H}{\partial \xi^i} \tag{1.9}$$

$$\frac{\partial L}{\partial \dot{\xi}^i} = -a_i \tag{1.10}$$

So we have:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\xi}^i}\right) - \frac{\partial L}{\partial \xi^i} &= -\frac{\partial a_i}{\partial \xi^j}\dot{\xi}^j + \frac{\partial a_j}{\partial \xi^i}\dot{\xi}^j + \frac{\partial H}{\partial \xi^i} \\ &= \left(\frac{\partial a_j}{\partial \xi^i} - \frac{\partial a_i}{\partial \xi^j}\right)\dot{\xi}^j + \frac{\partial H}{\partial \xi^i} = 0 \end{aligned} \tag{1.11}$$

Here we define the term $\frac{\partial a_j}{\partial \xi^i} - \frac{\partial a_i}{\partial \xi^j}$ as b_{ij}, then we have:

$$b_{ij}\dot{\xi}^j = -\frac{\partial H}{\partial \xi^i} \tag{1.12}$$

Here the $\boldsymbol{a}(\boldsymbol{\xi})$ is the Berry connection, while the b_{ij} is the Berry curvature.

For quantum system, here we only consider the coherent state, which means that the quantum state can be uniquely determined by a set of parameters, which is $|\psi(\boldsymbol{\xi})\rangle$. In this case the single point in phase space is exactly a quantum state. The Schrodinger equation is:

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = \hat{H}|\psi\rangle \tag{1.13}$$

Which is obtained from the definition of action:

$$S = \int_{t_1}^{t_2} \langle\psi| i\frac{\mathrm{d}}{\mathrm{d}t} - \hat{H}|\psi\rangle \tag{1.14}$$

And we can get:

$$i\langle\psi|\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle - \langle\psi|\hat{H}|\psi\rangle = 0 \tag{1.15}$$

Comparing with the generalized Lagrange equation, we have:

$$i\langle\psi|\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = -a_i(\xi)\dot{\xi}^i \tag{1.16}$$

Now look into the term $i\langle\psi|\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle$, which is:

$$i\langle\psi|\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = i\langle\psi|\frac{\partial}{\partial \xi^i}\dot{\xi}^i|\psi\rangle \tag{1.17}$$

$\dot{\xi}^i$ and $|\psi\rangle$ are commutable, so we have:

$$i\langle\psi|\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = i\langle\psi|\frac{\partial}{\partial \xi^i}|\psi\rangle\dot{\xi}^i \tag{1.18}$$

Thus we have:

$$a_i = -i\langle\psi|\frac{\partial}{\partial \xi^i}|\psi\rangle \tag{1.19}$$